

Task 1 — Voice Agent Architecture

How It Works (End to End)

Caller (Browser/Mic) → Whisper STT → LangGraph Agent (Ollama LLM) → Tools/RAG → Local TTS → A

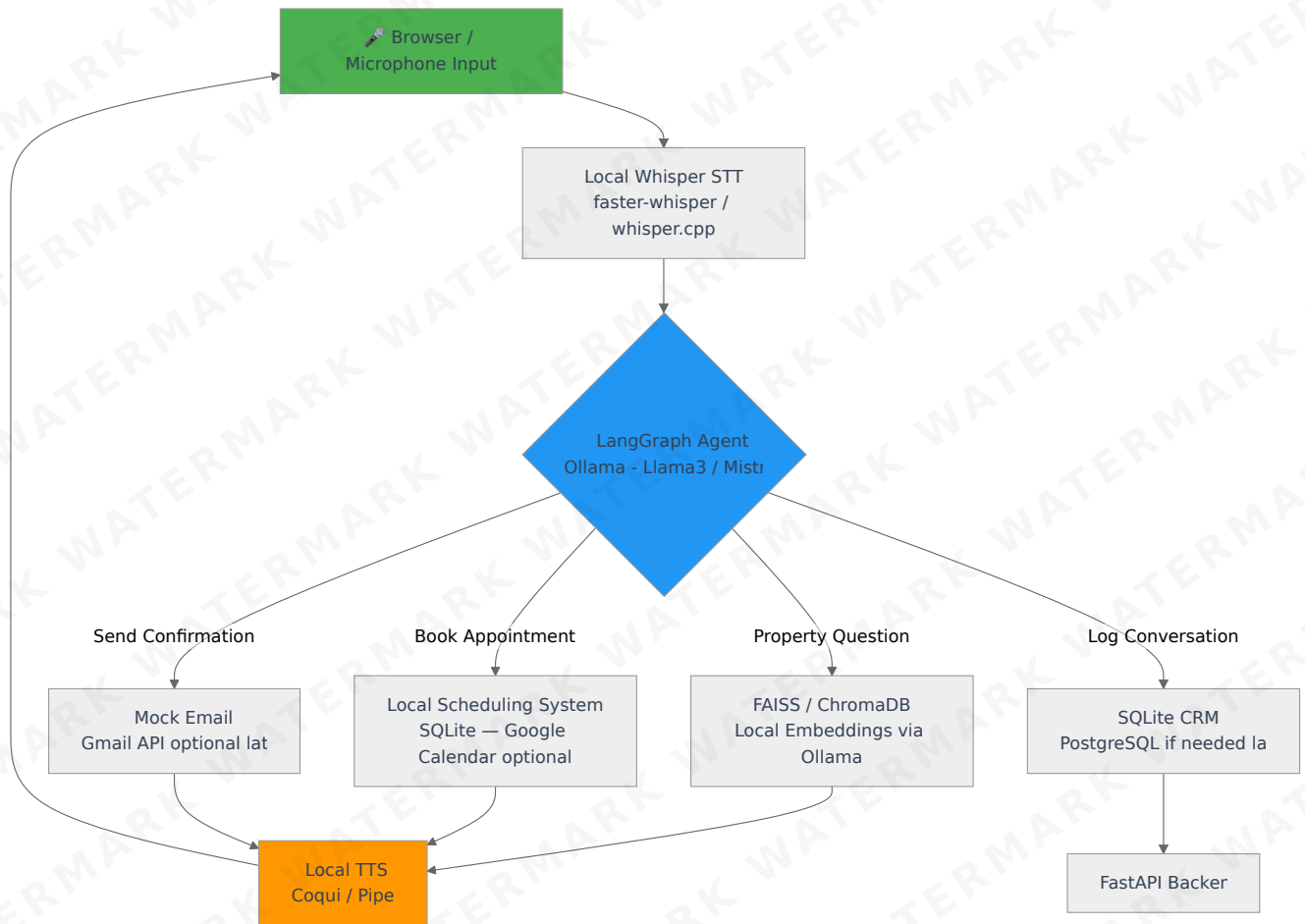
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FAISS/Chroma | SQLite | Local Calendar | Mock Email

Every call follows this pipeline:

1. **Telephony (Local Demo)** — Voice input from browser microphone. Real phone integration added later via Twilio or SIP.
2. **STT** — Local Whisper (`whisper.cpp` or `faster-whisper`) transcribes speech offline, no API cost.
3. **LLM** — Ollama runs a local open-source model (Llama 3.2 / Mistral / Qwen2.5). No OpenAI credits needed.
4. **Agent** — LangGraph orchestrates multi-step reasoning: understand intent → call tools → generate response.
5. **RAG** — Local embedding model (nomic-embed-text via Ollama) + FAISS/Chroma stores property listings and FAQs. All runs offline.
6. **Tools** — SQLite database for structured CRM data. Local scheduling system for appointments (Google Calendar added optionally later). Mock email initially, Gmail API added later if credentials are available.
7. **TTS** — Local TTS engine (Coqui TTS or Piper) converts response to audio. ElevenLabs free tier used only for demo comparison.
8. **Memory** — Conversation state stored per session in LangGraph state. Persisted to SQLite for returning customers.

Architecture Diagram



Technology Stack

Layer	Tool	Notes
LLM	Ollama (Llama 3.2 / Mistral / Qwen2.5)	Fully local, no API cost
Agent Framework	LangGraph + LangChain	Stateful workflows, tool calling
Embeddings	nomic-embed-text via Ollama	Local, no API needed
Vector DB	FAISS → ChromaDB	FAISS for speed, Chroma for persistence
Structured DB	SQLite → PostgreSQL	SQLite to start, scale later
STT	faster-whisper (local)	Offline transcription, fast

Layer	Tool	Notes
TTS	Coqui TTS / Piper (local)	ElevenLabs free tier for demo comparison only
Backend	FastAPI	Async, WebSocket for streaming
Email	Mock (print/log) → Gmail API	Free Gmail credentials added when available
Calendar	Local SQLite scheduler → Google Calendar	Optional GCal integration later
Telephony	Browser mic (local demo) → Twilio/SIP	Real phone integration added in later phase
Deployment	Local initially → Docker → Railway/Render	Progressive deployment

Progressive Build Strategy

Phase 1 (This Week): 100% local – Whisper + Ollama + FAISS + SQLite + local TTS + browser mic
Phase 2 (Optional): Add Gmail API, Google Calendar, ElevenLabs demo comparison
Phase 3 (Production): Add Twilio telephony, PostgreSQL, Docker, cloud deploy

This keeps Day 1–5 completely free to run with zero API costs.